



JIMMA INSTITUTE OF TECHNOLOGY
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Big data analytics and
Business intelligence project

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1. Introduction

This project focuses on performing Exploratory Data Analysis (EDA) on the Iris dataset using R. The purpose of this project is to understand the dataset structure, calculate summary statistics, perform data manipulation, apply logical conditions, and create visualizations to analyze relationships between variables.

2. Exploratory Data Analysis (EDA)

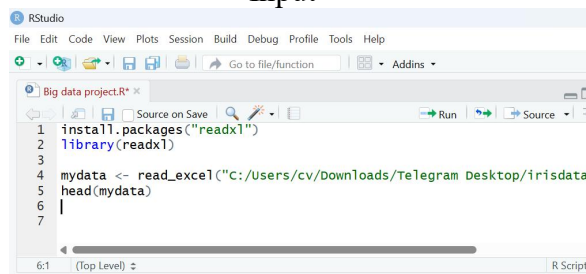
Question 1: Basic Information

a) Column names and dataset size

The column names of the dataset are: **Sepal.Length**, **Sepal.Width**, **Petal.Length**, **Petal.Width**, **Species**, **Id**.

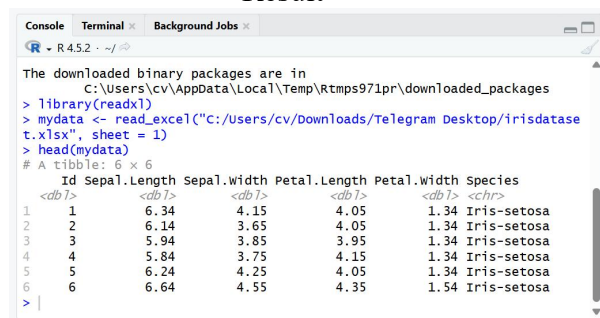
The dataset contains **150 rows** and **6 columns**.

Input



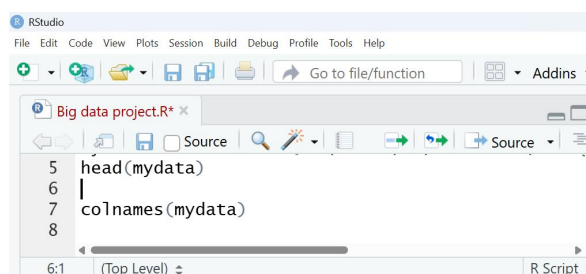
```
1 install.packages("readxl")
2 library(readxl)
3
4 mydata <- read_excel("C:/Users/cv/Downloads/Telegram Desktop/irisdata
5 head(mydata)
6
7
```

Result

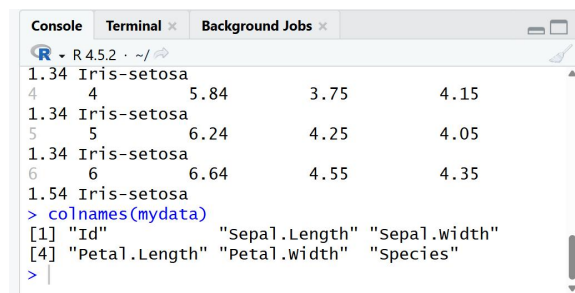


```
The downloaded binary packages are in
C:/Users/cv/AppData/Local/Temp/Rtmps971pr/downloaded_packages
> library(readxl)
> mydata <- read_excel("C:/Users/cv/Downloads/Telegram Desktop/irisdata
t.xlsx", sheet = 1)
> head(mydata)
# A tibble: 6 x 6
  Id Sepal.Length Sepal.Width Petal.Length Petal.Width Species
<dbl> <dbl> <dbl> <dbl> <dbl> <chr>
1 1 6.34 4.15 4.05 1.34 Iris-setosa
2 2 6.14 3.65 4.05 1.34 Iris-setosa
3 3 5.94 3.85 3.95 1.34 Iris-setosa
4 4 5.84 3.75 4.15 1.34 Iris-setosa
5 5 6.24 4.25 4.05 1.34 Iris-setosa
6 6 6.64 4.55 4.35 1.54 Iris-setosa
```

Input



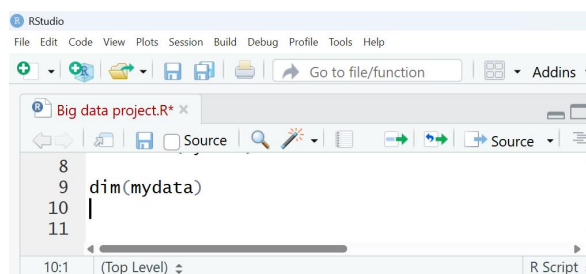
```
5 head(mydata)
6
7 colnames(mydata)
8
```



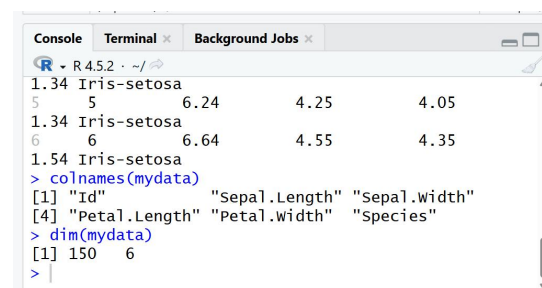
```
1.34 Iris-setosa
4 4 5.84 3.75 4.15
1.34 Iris-setosa
5 5 6.24 4.25 4.05
1.34 Iris-setosa
6 6 6.64 4.55 4.35
1.54 Iris-setosa
> colnames(mydata)
[1] "Id" "Sepal.Length" "Sepal.Width"
[4] "Petal.Length" "Petal.Width" "Species"
```

Result Shows the columns in the dataset

Input



```
8
9 dim(mydata)
10
11
```



```
1.34 Iris-setosa
5 5 6.24 4.25 4.05
1.34 Iris-setosa
6 6 6.64 4.55 4.35
1.54 Iris-setosa
> colnames(mydata)
[1] "Id" "Sepal.Length" "Sepal.Width"
[4] "Petal.Length" "Petal.Width" "Species"
> dim(mydata)
[1] 150 6
```

Result - Shows the number of rows and column

b) Missing values

Missing values were checked using R functions. The results show that **there are no missing values** in the dataset. All columns contain complete data.

Input

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Big data project.R*
10
11 sum(is.na(mydata))
12
13
14
11:19 (Top Level) R Script
```

Result

```
R - R 4.5.2 - ~/
1.34 Iris-setosa
6      6      6.64      4.55      4.35
1.54 Iris-setosa
> colnames(mydata)
[1] "Id"      "Sepal.Length" "Sepal.Width"
[4] "Petal.Length" "Petal.Width"  "Species"
> dim(mydata)
[1] 150 6
> sum(is.na(mydata))
[1] 0
>
>
```

Input

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Big data project.R*
10
11 sum(is.na(mydata))
12 colSums(is.na(mydata))
13
14
12:23 (Top Level) R Script
```

Result

```
R - R 4.5.2 - ~/
[4] "Petal.Length" "Petal.Width" "Species"
> dim(mydata)
[1] 150 6
> sum(is.na(mydata))
[1] 0
> colSums(is.na(mydata))
      Id Sepal.Length Sepal.Width
Petal.Length Petal.Width Species
      0           0           0
      0           0           0
>
>
```

Question 2: Summary Statistics

The mean, median, and standard deviation were calculated for the numeric columns:

- Sepal.Length
- Sepal.Width
- Petal.Length
- Petal.Width

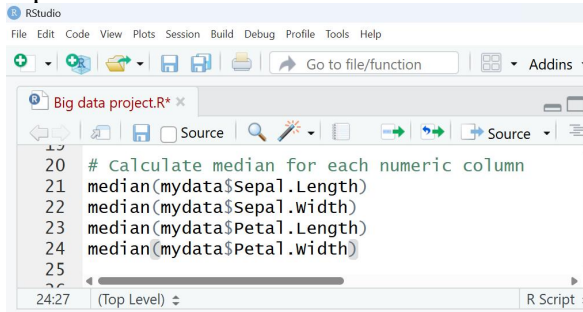
Input

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Big data project.R*
14 # Calculate mean for each numeric column
15 mean(mydata$Sepal.Length)
16 mean(mydata$Sepal.Width)
17 mean(mydata$Petal.Length)
18 mean(mydata$Petal.Width)
19
18:25 (Top Level) R Script
```

Result

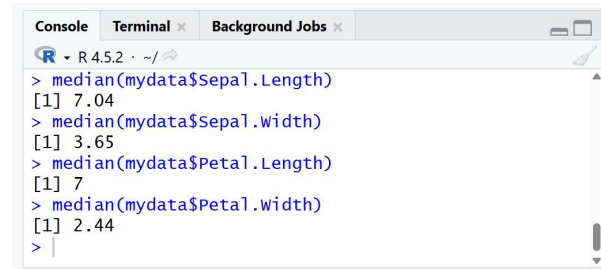
```
R - R 4.5.2 - ~/
> mean(mydata$Sepal.Length)
[1] 7.083333
> mean(mydata$Sepal.Width)
[1] 3.704
> mean(mydata$Petal.Length)
[1] 6.408667
> mean(mydata$Petal.Width)
[1] 2.338667
>
>
```

Input



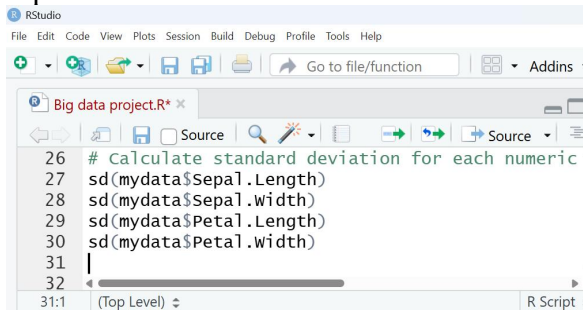
```
20 # Calculate median for each numeric column
21 median(mydata$Sepal.Length)
22 median(mydata$Sepal.Width)
23 median(mydata$Petal.Length)
24 median(mydata$Petal.Width)
```

Result



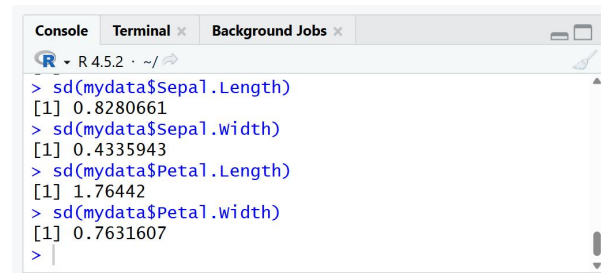
```
> median(mydata$Sepal.Length)
[1] 7.04
> median(mydata$Sepal.Width)
[1] 3.65
> median(mydata$Petal.Length)
[1] 7
> median(mydata$Petal.Width)
[1] 2.44
```

Input



```
26 # Calculate standard deviation for each numeric
27 sd(mydata$Sepal.Length)
28 sd(mydata$Sepal.Width)
29 sd(mydata$Petal.Length)
30 sd(mydata$Petal.Width)
```

Result



```
> sd(mydata$Sepal.Length)
[1] 0.8280661
> sd(mydata$Sepal.Width)
[1] 0.4335943
> sd(mydata$Petal.Length)
[1] 1.76442
> sd(mydata$Petal.Width)
[1] 0.7631607
```

These statistics help understand the central tendency and variability of the data.

After comparing the standard deviation values, Petal.Length has the highest variation, meaning it is the most spread-out variable.

Question 3: Visualization

A scatter plot was created to show the relationship between **Sepal.Length** and **Sepal.Width**, colored by species.

Observations:

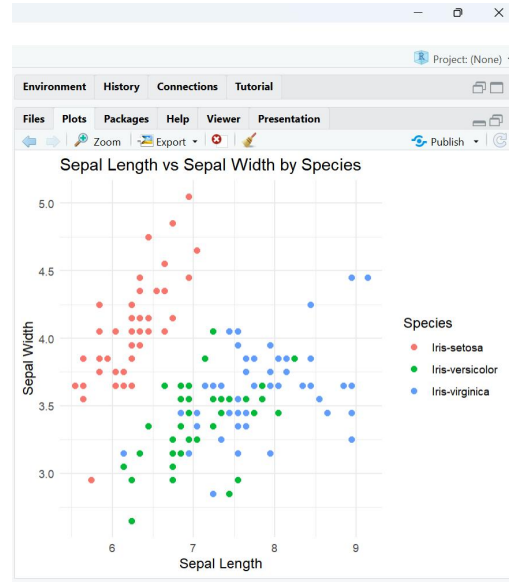
- Iris-setosa forms a clearly separate cluster.
- Iris-versicolor and Iris-virginica overlap slightly.
- The relationship between sepal length and width varies across species.

The species Iris-setosa shows a strong positive correlation between sepal length and width, indicated by a tight, narrow clustering of data points. This suggests a highly predictable biological relationship. In contrast, Iris-versicolor and Iris-virginica exhibit a more moderate correlation; their data points are more scattered (wider 'clouds'), indicating that their sepal dimensions vary more independently of one another.

Input

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Big data project.R*
ggplot(data = mydata , aes(x = Sepal.Length, y = Sepal.Width,
53 color = Species)) +
54 geom_point(size = 2) +
55 theme_minimal() +
56 labs(
57 title = "Sepal Length vs Sepal width by Species",
58 x = "Sepal Length",
59 y = "Sepal width"
60 )
61
62
63
60:22 (Top Level) R Script
```

Result



3. Data manipulation

Question 4: Using dplyr

a) Selecting specific columns

Only the columns **Sepal.Length**, **Sepal.Width**, and **Species** were selected, and the first 10 rows were displayed.

Input

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Big data project.R*
library(dplyr)
63
64
65 # select specific columns from the iris dataset
66 selected_data <- mydata %>%
67   select(Sepal.Length, Sepal.Width, Species)
68
69 # Display the first 10 rows of the selected dataset
70 head(selected_data, 10)
71
72
73
71:1 (Top Level) R Script
```

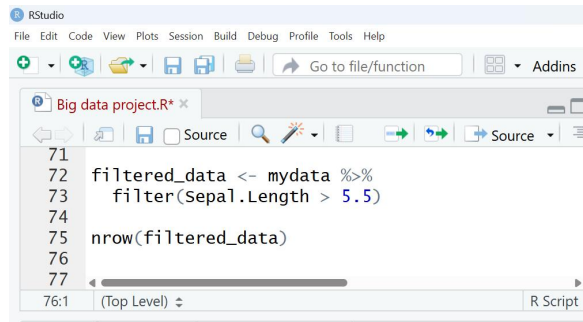
Result

```
R - R452 - ~/
+ select(Sepal.Length, Sepal.Width, Species)
> head(selected_data, 10)
# A tibble: 10 x 3
  Sepal.Length Sepal.Width Species
  <dbl>         <dbl>   <chr>
1      6.34         4.15 Iris-setosa
2      6.14         3.65 Iris-setosa
3      5.94         3.85 Iris-setosa
4      5.84         3.75 Iris-setosa
5      6.24         4.25 Iris-setosa
6      6.64         4.55 Iris-setosa
7      5.84         4.05 Iris-setosa
8      6.24         4.05 Iris-setosa
9      5.64         3.55 Iris-setosa
10     6.14         3.75 Iris-setosa
```

b) Filtering rows

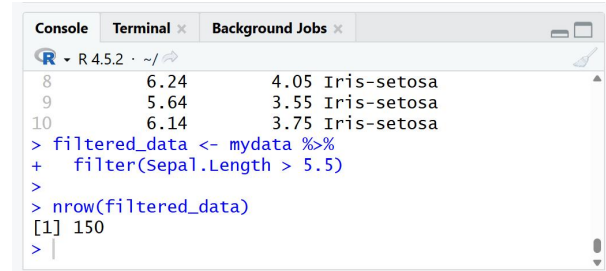
The dataset was filtered to show only rows where **Sepal.Length** is greater than **5.5**. The number of rows meeting this condition was calculated using `nrow()` (and it is 150).

Input



```
71
72 filtered_data <- mydata %>%
73   filter(Sepal.Length > 5.5)
74
75 nrow(filtered_data)
76
77
```

Result



```
R - R4.5.2 - ~/
8      6.24      4.05 Iris-setosa
9      5.64      3.55 Iris-setosa
10     6.14      3.75 Iris-setosa
> filtered_data <- mydata %>%
+   filter(Sepal.Length > 5.5)
>
> nrow(filtered_data)
[1] 150
>
```

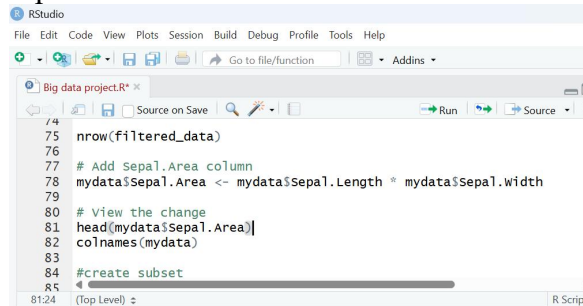
Question 5: Add a new column

A new column named **Sepal.Area** was added to the dataset. It was calculated as:

$$\text{Sepal.Area} = \text{Sepal.Length} \times \text{Sepal.Width}$$

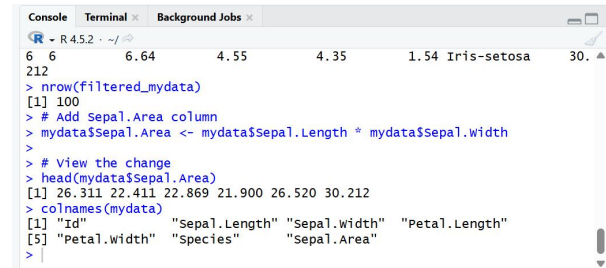
This new column provides additional information about the sepal size.

Input



```
74
75 nrow(filtered_data)
76
77 # Add Sepal.Area column
78 mydata$Sepal.Area <- mydata$Sepal.Length * mydata$Sepal.Width
79
80 # View the change
81 head(mydata$Sepal.Area)
82 colnames(mydata)
83
84 #create subset
85
```

Result



```
R - R4.5.2 - ~/
6      6      6.64      4.55      4.35      1.54 Iris-setosa      30.
212
> nrow(filtered_mydata)
[1] 100
> # Add Sepal.Area column
> mydata$Sepal.Area <- mydata$Sepal.Length * mydata$Sepal.Width
>
> # View the change
> head(mydata$Sepal.Area)
[1] 26.311 22.411 22.869 21.900 26.520 30.212
> colnames(mydata)
[1] "Id"      "Sepal.Length" "Sepal.Width" "Petal.Length"
[5] "Petal.Width" "Species"      "Sepal.Area"
>
```

4. Data Subsetting Using Logical Operators

Question 6: Subset Based on Sepal Measurements

A subset was created where:

- `Sepal.Length > 5.0`
- `Sepal.Width < 3.0`

The number of rows satisfying this condition was calculated.

Input

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Source on Save Run Go to file/function Addins
Big data project.R
82 colnames(mydata)
83
84 #create subset
85 subset_mydata <- mydata[mydata$Sepal.Length > 5.0
86 & mydata$Sepal.Width < 3.0, ]
87
88 head(subset_mydata)
89 #Number of rows
90 nrow(subset_mydata)
91
92 subset_mydata2 <- mydata[
93 (mydata$Species == "Iris-setosa"
94
91:1 (Top Level) R Script
```

Result

```
R - R 4.5.2 - ~/
Console Terminal Background Jobs
> subset_mydata <- mydata[mydata$Sepal.Length > 5.0
+ & mydata$Sepal.Width < 3.0, ]
> head(subset_mydata)
# A tibble: 6 x 7
  Id Sepal.Length Sepal.Width Petal.Length Petal.Width Species Sepal.Area
<dbl> <dbl> <dbl> <dbl> <dbl> <chr> <dbl>
1 42 5.74 2.95 3.95 1.44 Iris-seto... 16.9
2 54 6.74 2.95 6.65 2.44 Iris-vers... 19.9
3 61 6.24 2.65 6.15 2.14 Iris-vers... 16.5
4 63 7.24 2.85 6.65 2.14 Iris-vers... 20.6
5 69 7.44 2.85 7.15 2.64 Iris-vers... 21.2
6 88 7.54 2.95 7.05 2.44 Iris-vers... 22.2
> #Number of rows
> nrow(subset_mydata)
[1] 8
```

Question 7: AND and OR Logical Conditions

A subset was created where:

- Species is either **Iris-setosa** or **Iris-versicolor**
- Petal.Length > 2.0

Logical operators AND (&) and OR (|) were used together to apply these conditions.

Input

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Source on Save Run Go to file/function Addins
Big data project.R
91
92 subset_mydata2 <- mydata[
93 (mydata$Species == "Iris-setosa"
94 | mydata$Species == "Iris-versicolor") &
95 mydata$Petal.Length > 2.0,
96 ]
97
98 head(subset_mydata2)
99 #Number of rows
100 nrow(subset_mydata2)
101
101:1 (Top Level) R Script
```

Result

```
R - R 4.5.2 - ~/
Console Terminal Background Jobs
> head(subset_mydata2)
# A tibble: 6 x 7
  Id Sepal.Length Sepal.Width Petal.Length Petal.Width Species Sepal.Area
<dbl> <dbl> <dbl> <dbl> <dbl> <chr> <dbl>
1 42 5.74 2.95 3.95 1.44 Iris-setosa 16.9
2 54 6.74 2.95 6.65 2.44 Iris-versicolor 19.9
3 61 6.24 2.65 6.15 2.14 Iris-versicolor 16.5
4 63 7.24 2.85 6.65 2.14 Iris-versicolor 20.6
5 69 7.44 2.85 7.15 2.64 Iris-versicolor 21.2
6 88 7.54 2.95 7.05 2.44 Iris-versicolor 22.2
> #Number of rows
> nrow(subset_mydata2)
[1] 100
```

Question 8: Excluding Rows

Rows where:

- Species is **Iris-virginica**
- Sepal.Length > 6.0

were excluded from the dataset. The remaining number of rows was calculated after exclusion

Input

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Source on Save Run Go to file/function Addins
Big data project.R
97
98 head(subset_mydata2)
99 #Number of rows
100 nrow(subset_mydata2)
101
102
103 filtered_mydata <- mydata[!(mydata$Species == "Iris-virginica"
104 & mydata$Sepal.Length > 6.0),]
105 head(filtered_mydata)
106 nrow(filtered_mydata)
107
107:1 (Top Level) R Script
```

Result

```
R - R 4.5.2 - ~/
Console Terminal Background Jobs
> nrow(filtered_mydata)
[1] 100
> filtered_mydata <- mydata[!(mydata$Species == "Iris-virginica"
+ & mydata$Sepal.Length > 6.0),]
> head(filtered_mydata)
# A tibble: 6 x 7
  Id Sepal.Length Sepal.Width Petal.Length Petal.Width Species Sepal.Area
<dbl> <dbl> <dbl> <dbl> <dbl> <chr> <dbl>
1 1 6.34 4.15 4.05 1.34 Iris-setosa 26.311
2 2 6.14 3.65 4.05 1.34 Iris-setosa 22.411
3 3 5.94 3.85 3.95 1.34 Iris-setosa 22.869
4 4 5.84 3.75 4.15 1.34 Iris-setosa 21.900
5 5 6.24 4.25 4.05 1.34 Iris-setosa 26.520
6 6 6.64 4.55 4.35 1.54 Iris-setosa 30.212
> nrow(filtered_mydata)
[1] 100
>
```